

BADER AL QASEM

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EDUCATION

University of Alberta
Bachelor of Science in Computing Science (Specialization)

Edmonton, AB
Expected Dec 2026

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), TypeScript, Python, C, SQL

Frameworks & APIs: React, Node.js, Express, FastAPI, REST APIs, JWT, OpenAI API

Databases & DevOps: PostgreSQL, Docker, Git, GitHub Actions (CI/CD), Azure

Testing: Unit Testing, JUnit, Postman

EXPERIENCE

Software Developer (Full Stack)
Next Up Hoops ([Live Site](#)) | [GitHub](#) | [Demo \(Loom\)](#)

Edmonton, AB
Nov 2024 – Present

- Designed, built, and deployed a **full-stack platform** for a **120+ athlete** youth basketball organization.
- Developed backend using **Node.js, Express, PostgreSQL**, implementing **JWT-based authentication** to support **role-based access** for admins and players.
- Implemented registration, tryout, payment, and account-management workflows, reducing administrative processing time by over **80%**.
- Integrated **Stripe Checkout**, automated email flows, and password recovery; wrote **Postman API tests** to validate authentication, registration, and payment flows.

PROJECTS

Custom Memory Allocator | [GitHub](#)

- Implemented a custom **malloc/free memory allocator** in **C**, supporting heap expansion, block splitting, and coalescing.
- Designed **dual linked-list structures** (heap list and free list) to optimize memory reuse and allocation efficiency.
- Built randomized **stress tests** to validate correctness under memory fragmentation.

Resume Skill Gap Analyzer | [GitHub](#)

- Built a resume-job matching system using **Sentence-BERT embeddings** and the **OpenAI API** to identify skill gaps and suggest relevant learning resources.
- Designed and deployed containerized **FastAPI services** to **Azure** with automated **CI/CD** via GitHub Actions.
- Implemented REST APIs for embedding comparison and feedback generation.

Event Lottery App | [GitHub](#)

- Collaborated in a **6-person Agile/Scrum team** to develop an **Android application in Java** using **Firebase** supporting Entrant, Organizer, and Admin roles.
- Followed industry-standard workflows including **user-story-driven issues**, **sprint planning**, and **peer code reviews** via GitHub Projects.
- Implemented **JUnit unit tests** and **Espresso UI tests** to validate user flows and ensure application reliability.